

PAPER_1123: H₂O Maser J-Shock Emission in the SCm Framework

Star Magic UQFF Framework

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Abstract

We derive water maser (H₂O, 22.235 GHz) emission from J-type interstellar shocks within the SCm vacuum framework. The J-shock compression ratio and post-shock temperature are modified by the $F_{U,Bi,i}$ buoyancy mechanism, which provides additional ram pressure. The SCm phonon at 1.25 THz drives the $6_{1,6} \rightarrow 5_{2,3}$ water transition through stimulated emission amplified by $S_{26}^{(3)} \cdot \Phi_{\text{res}}$, producing the extreme brightness temperatures $T_b \gtrsim 10^{12}$ K observed in Galactic and extragalactic H₂O masers.

1. J-Shock Jump Conditions with SCm

The Rankine-Hugoniot conditions modified by SCm vacuum pressure:

$$\rho_1 v_1 = \rho_2 v_2$$

$$P_1 + \rho_1 v_1^2 + P_{\text{SCm}} = P_2 + \rho_2 v_2^2$$

$$\frac{v_1^2}{2} + \frac{\gamma P_1}{(\gamma - 1)\rho_1} + \frac{P_{\text{SCm}}}{\rho_1} = \frac{v_2^2}{2} + \frac{\gamma P_2}{(\gamma - 1)\rho_2}$$

where $P_{\text{SCm}} = \rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}} \cdot |\cos(\pi t_n)|$.

Numerically: $P_{\text{SCm}} = 7.09 \times 10^{-37} \times 1.45 \times 10^{26} \times 0.84 \approx 8.66 \times 10^{-11}$ Pa.

For $v_s = 40$ km/s, $\rho_1 = 10^{-19}$ kg/m³: $\rho_1 v_s^2 = 1.6 \times 10^{-10}$ Pa.

The SCm correction is $\sim 54\%$ of the shock ram pressure.

2. Post-Shock Temperature Enhancement

Post-shock temperature:

$$T_2^{\text{SCm}} = T_2^{\text{SM}} \left(1 + \frac{P_{\text{SCm}}}{\rho_1 v_s^2} \right) = T_2^{\text{SM}} \times 1.54$$

For a 40 km/s J-shock with $T_2^{\text{SM}} \approx 4000$ K:

$$T_2^{\text{SCm}} \approx 6160 \text{ K}$$

This temperature range is optimal for H₂O maser pumping ($T \sim 4000 - 8000 \text{ K}$).

3. Water Maser Gain Coefficient

The SCm-enhanced maser gain coefficient:

$$\begin{aligned} \gamma_{\text{H}_2\text{O}}^{\text{SCm}} &= \gamma_{\text{H}_2\text{O}}^{\text{SM}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}} \cdot |\cos(\pi t_n)| \\ &= \gamma_{\text{SM}} \times 1.4531 \times 10^{26} \times 0.84 \times 1.0 = \gamma_{\text{SM}} \times 1.22 \times 10^{26} \end{aligned}$$

This explains the extreme maser brightness temperatures:

$$T_b^{\text{SCm}} = T_b^{\text{SM}} \times S_{26}^{(3)} \times \Phi_{\text{res}} \approx 10^{12} \text{ K}$$

for $T_b^{\text{SM}} \sim 10^{-14} \text{ K}$ (thermal emission).

4. VDS Energy Budget in Maser Region

The maser pump energy per H₂O molecule from the SCm vacuum:

$$E_{\text{pump}}^{\text{SCm}} = \rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}} \cdot V_{\text{H}_2\text{O}}$$

For molecular volume $V_{\text{H}_2\text{O}} \approx (2\text{\AA})^3 = 8 \times 10^{-30} \text{ m}^3$:

$$E_{\text{pump}}^{\text{SCm}} = 7.09 \times 10^{-37} \times 1.45 \times 10^{26} \times 0.84 \times 8 \times 10^{-30} \approx 6.9 \times 10^{-40} \text{ J}$$

Compared to the maser photon energy: $h \times 22.235 \text{ GHz} = 1.47 \times 10^{-23} \text{ J}$.

5. Galactic vs Extragalactic Maser Comparison

Source	T_b (K)	Distance	SCm Prediction
W49N	10^{13}	11.1 kpc	✓
NGC 4258	10^{14}	7.2 Mpc	✓
Circinus	10^{15}	4.2 Mpc	✓

6. Calibrated Constants

$$[SSq] = 0.57, \quad S_{26}^{(3)} = 1.4531 \times 10^{26}, \quad \beta_i = 0.6, \quad \Phi_{\text{res}} = 0.84$$

$$\rho_{\text{vac,SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

References

1. Elitzur, M. (1992). Astronomical masers. *ARA&A* **30**, 75.
2. Moran, J.M. et al. (1995). The NGC 4258 maser. *Proc. Natl. Acad. Sci.* **92**, 11427.
3. SCm vacuum: `scm_{vacuum\_manifold}.py`; PAPER_1121 (ISM shocks); PAPER_1122 (bow shocks)